

Math211

Discrete Mathematics

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Lecture 1

Propositional Logic I



Course Grading Schema

Total score 100 degrees

Category	Number	Percentage
Final exam	1	30
Midterm	1	30
Quizzes	3 (best 2)	20
Assignments	3 (average)	7
Lec. Contribution	7	5
Tutorial Contribution		8
Project	3	<i>(+3 marks bonus)</i>
Lec. & Tut. Bonus	2	<i>(+2 marks bonus)</i>

Percentages are subject to changes depending on circumstances at the time

Class policy



- ▶ Lecture notes will be posted on Moodle



Assignment Policy

- ▶ Assignments will be posted on Moodle
 - ▶ **No late submissions**
- ▶ All assignments must be done **individually**
- ▶ Cheating:
 - ▶ Helping others, getting help, looking up website for solutions, etc.
- ▶ Students caught cheating will be awarded an **F** grade, and will be subject to the NU academic dishonesty policy.
- ▶ If something is not clear, on what constitutes and what does not, please consult the TA in advance.



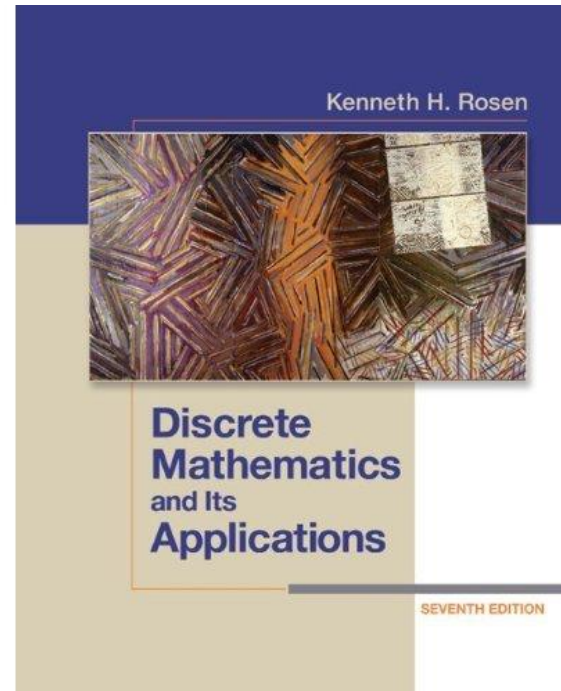
Topics

- ▶ Logic
- ▶ Proof Techniques
- ▶ Sets
- ▶ Functions
- ▶ Matrices and Integers
- ▶ Relations
- ▶ Graph



Textbook

Discrete Mathematics and Its Applications by Kenneth H. Rosen,
12th edition, McGraw Hill



Why Study Discrete Math?

The basis of all digital information processing:

- ▶ Mathematics is at the heart of *problem solving*.
- ▶ Computers use *discrete structures* to represent and manipulate data.
- ▶ Discrete manipulations of discrete structures represented in memory.
- ▶ It's the basic language and conceptual foundation of all of computer science.
- ▶ Discrete concepts are also widely used throughout math, science, engineering, economics, biology, etc., ...



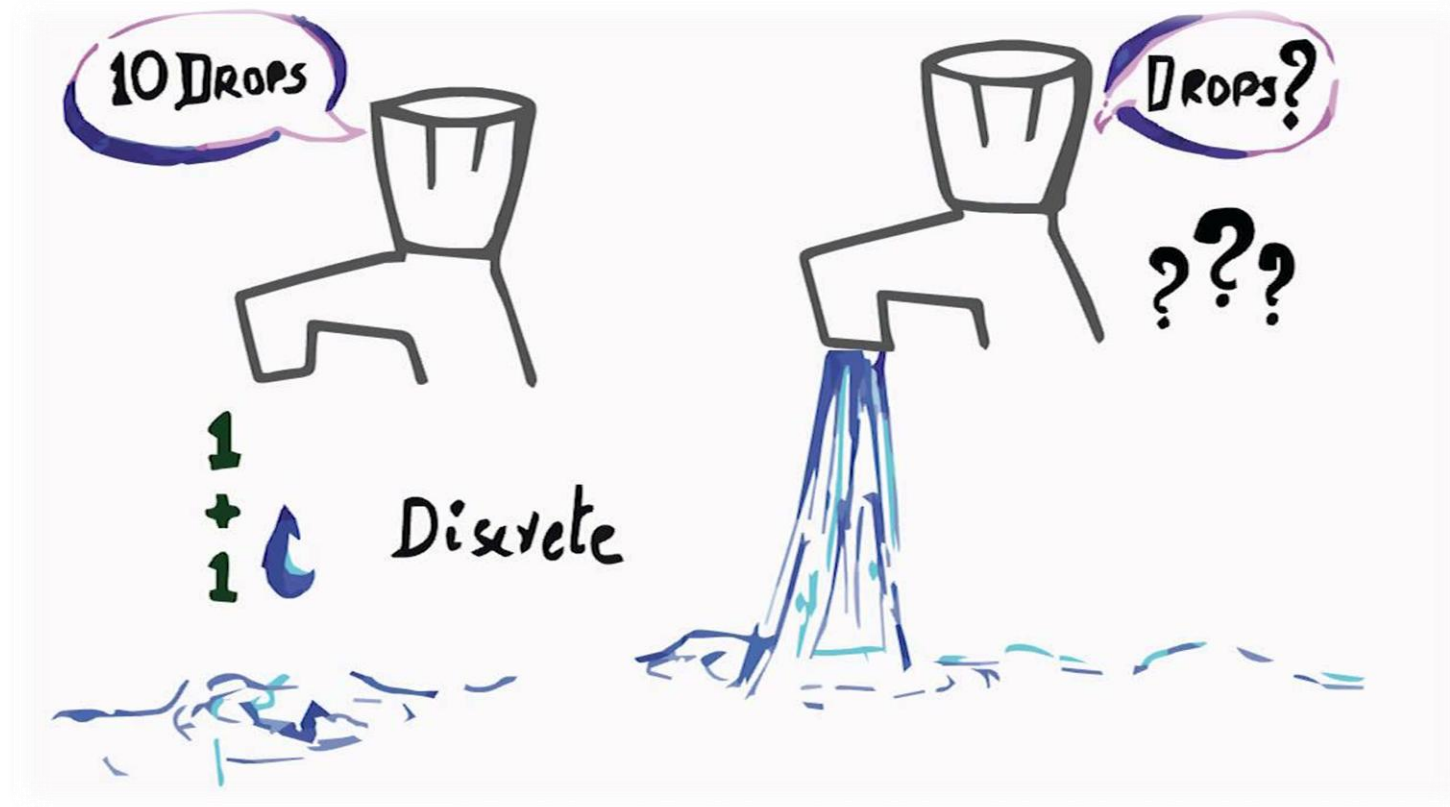
Applications of Discrete Math in CS

- ▶ Advanced algorithms & data structures
- ▶ Programming language compilers & interpreters
- ▶ Computer networks
- ▶ Operating systems
- ▶ Computer architecture
- ▶ Relational Database
- ▶ Cryptography
- ▶ Error correction codes
- ▶ Graphics & animation algorithms, game engines
- ▶ Automata Theory



The whole field



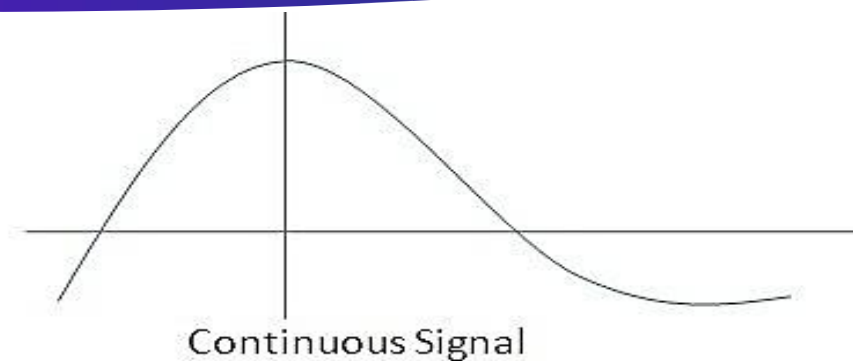


Continuous versus Discrete Math

Continuous versus Discrete

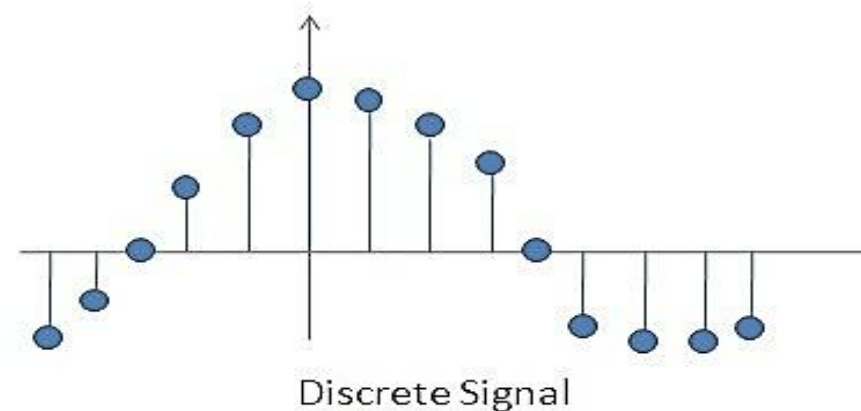
Examples of continuous Data

- ▶ Height of a person
- ▶ Time in a race
- ▶ Distance traveled by a car



Examples of discrete Data

- ▶ Number of boys in the class
- ▶ Number of candies in a packet
- ▶ Number of suitcases lost by an airline



1.1 Propositional Logic

- Proposition
- Logical Connectives
- Conditional Statements
- Truth Tables
- Compound Propositions
- Precedence of Logical Operators

Proposition

A declarative sentence that is either *true* or *false*, but not both

- ▶ “Today is Friday”
 - ▶ It is not the case that today is Friday
 - ▶ Today is not Friday

- ▶ Cairo is the capital of Egypt
- ▶ Alexandria is adjacent to Aswan
- ▶ $1+1=2$
- ▶ $2+2=5$

The proposition is TRUE (T)

The proposition is FALSE (F)

The proposition is TRUE (T)

The proposition is FALSE (F)



Proposition

1. What time is it?
2. Read this carefully.
3. $x + 1 = 2$.
4. $x + y = z$.

- ▶ **Sentences 1 and 2 are not propositions** because they are not declarative sentences.
- ▶ **Sentences 3 and 4 are not propositions** because they are neither true nor false.
- ▶ Note that each of sentences 3 and 4 can be turned into a proposition if we assign values to the variables



Example

$x+2 \geq y$ is a sentence but not a proposition;

- ▶ $x^2+y > \{ \}$ is not a sentence
- ▶ $x+2 \geq y$ is true if the number $x+2$ is not less than the number y
- ▶ $x+2 \geq y$ is true in a world where $x = 7, y = 1$
- ▶ $x+2 \geq y$ is false in a world where $x = 0, y = 6$



Notations

- ▶ The small letters are commonly used to denote the **propositional variables**, that is, variables that represent propositions, such as, $p, q, r, s, \dots$
 - **Today is Friday:** p
 - **It is raining today:** q
- ▶ The truth value of a proposition is **true**, denoted by T or 1 , if it is a true proposition and **false**, denoted by F or 0 , if it is a false proposition.



Logical operators/connectives

Operators: unary and binary

- ▶ Negation operator ($\neg$)
- ▶ Conjunction (and, $\wedge$)
- ▶ Disjunction (or, $\vee$)
- ▶ Exclusive Or ($\oplus$)
- ▶ Conditional statement [implication/if-then] ($\rightarrow$)
- ▶ Biconditional statement [if and only if] ($\leftrightarrow$)



Compound statements

- ▶ Simple Propositions that are connected by logical operators are called “Compound statements”.
- ▶ p : Today is Friday.
- ▶ q : It is raining.
- ▶ $p \wedge q$: Today is Friday and it is raining.



Truth Table

- ▶ A truth table lists the results for each possible arrangement of operands
 - ▶ Order is important: $p \text{ op } q$ may or may not give the same result as $q \text{ op } p$
- ▶ The rows in a truth table list all possible sequences of truth values for n operands, and specify a result for each sequence
 - ▶ Hence, there are 2^n rows in a truth table for n operands



Negation (not)

Unary operator

$n = 1$ operand

$2^n = 2^1 = 2$ rows

p	$\neg p$
T	F
F	T



Example

Find the negation of the following proposition

p : Today is Friday.

The negation is

$\neg p$: It is not the case that today is Friday.

This negation can be more simply expressed by

$\neg p$: Today is not Friday.



Conjunction (and) ($\wedge$)

Conjunction (*AND*)

- ▶ $p \wedge q$ is true when **both** p and q are true
- ▶ False otherwise

$n = 2$ operands

$2^n = 2^2 = 4$ rows

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F



Example

- ▶ p : “Today is Friday”, q : “It is raining today”
- ▶ $p \wedge q$ “Today is Friday and it is raining today”
 - ▶ true: on rainy Fridays
 - ▶ false otherwise:
 - ▶ Any day that is not a Friday
 - ▶ Fridays when it does not rain



Disjunction (or) ($\vee$)

Disjunction (OR)

- ▶ $p \vee q$ is false when both p and q are false.
- ▶ True otherwise

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F



Example

- ▶ $p \vee q$: “Today is Friday or it is raining today”
 - ▶ True
 - ▶ Today is Friday
 - ▶ It is raining today
 - ▶ It is a rainy Friday
 - ▶ False
 - ▶ Today is not Friday and it does not rain



Exclusive Or

Exclusive OR (XOR)

- ▶ is true when exactly one of p or q is true.
- ▶ False otherwise

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F



Example

p : Samy will **pass** the Math211 course.

q : Samy will **fail** the Math211 course.

The exclusive or is

$p \oplus q$: Samy will **either pass or fail** the course Math211 but not both.



AND, OR, Exclusive OR

“Students who have **taken math and computer science** can take this class.”

Here, we mean that only students who have taken both calculus and computer science can take the class.

“Students who have **taken math or computer science** can take this class.”

Here, we mean that students who have taken both calculus and computer science can take the class, as well as the students who have taken only one of the two subjects.

“Students who have **taken math or computer science, but not both**, can enroll in this class.”

Here, we mean that students who have taken both calculus and a computer science course cannot take the class. Only those who have taken exactly one of the two courses can take the class.



Conditional statement

If p - then q (*implies*)

- ▶ p is the hypothesis/premise/antecedent
- ▶ q is the conclusion/consequent
- ▶ $p \rightarrow q$ is false when p is true, and q is false.
- ▶ True otherwise

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T



Conditional statement

- ▶ “if p , then q ”
- ▶ “ p implies q ”
- ▶ “if p , q ”
- ▶ “ p only if q ”
- ▶ “ p is sufficient for q ”
- ▶ “a sufficient condition for q is p ”
- ▶ “ q if p ”
- ▶ “ q whenever p ”
- ▶ “ q when p ”
- ▶ “ q is necessary for p ”
- ▶ “a necessary condition for p is q ”
- ▶ “ q follows from p ”



Example

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

"If you study hard, then you will pass the exam"

1. The student **studied** hard and **passed** the exam.
 ▶ is true. Note that p and q are both true.
2. The student **did not study** hard and **did not pass** the exam.
 ▶ is true. Note that p and q are both false.
3. The student **did not study** hard but **passed** the exam.
 ▶ is true. Note that p is false, but q is true.
4. The student **studied** hard and **didn't pass** the exam.
 ▶ is false. Note that p is true, but q is false.



Example

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

A person makes the promise: "If I find a L.E. 200 bill, then I will take you to a movie." Now we consider when the person has broken the promise.

1. The person **finds a L.E. 200** bill and **takes you** to a movie.
 - ▶ is true, since the person did not break the promise. Note p and q are both true.
2. The person **did not find** a L.E. 200 bill **but took you** to a movie anyway.
 - ▶ is true, since the person did not break the promise. Note p is false, but q is true.
3. The person **did not find** a L.E. 200 bill and **did not take** you to a movie.
 - ▶ is true, since the person did not break the promise. Note p and q are both false.
4. The person **finds** a L.E. 200 bill and **does not take** you to a movie.
 - ▶ is false, since the person broke the promise. Note p is true, but q is false.



Biconditional statement

If and only if

- ▶ “p if and only if q”
- ▶ $p \leftrightarrow q$ is true when p and q have the same truth value.
- ▶ False otherwise
- ▶ Also known as bi-implications

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T



Example

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

"You will pass if and only if you study hard."

1. The student **studied** hard and **passed**.
 ▶ is true. Note that p and q are both true.
2. The student **studied** hard and **did not pass**.
 ▶ is false. Note that p is true, but q is false.
3. The student **did not study** hard and **passed**.
 ▶ is false. Note that p is false, but q is true.
4. The student **did not study** hard and **did not pass**.
 ▶ is true. Note that p and q are both false.



Example

- ▶ p : “you can take the flight”, q : “you buy a ticket”
- ▶ $p \leftrightarrow q$: “You can take the flight *if and only if* you buy a ticket”
 - ▶ This statement is true
 - ▶ If you buy a ticket and take the flight
 - ▶ If you do not buy a ticket and you cannot take the flight



Precedence of logic operators

$$X = 2 + 3 \times 5$$

$$X = (2 + 3) \times 5$$

$$\neg p \wedge q$$

$$\neg (p \wedge q)$$

Operator	Symbol	Precedence
Brackets	()	1
Not	$\neg$	2
And	$\wedge$	3
OR	$\vee$	4
XOR	$\oplus$	5
Conditional	$\rightarrow$	6
Biconditional	$\leftrightarrow$	7



Truth table of compound propositions $(\neg p) \vee (\neg q)$

$$(\neg p) \vee (\neg q)$$

p	q	$\neg p$	$\neg q$	$(\neg p) \vee (\neg q)$
true	true	false	false	false
true	false	false	true	true
false	true	true	false	true
false	false	true	true	true



Truth table of compound propositions

$$(p \vee \neg q) \rightarrow (p \wedge q)$$



Truth table of compound propositions

$$(p \vee \neg q) \rightarrow (p \wedge q)$$

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F



Review Questions



$$a) (p \vee q) \vee r$$

$$b) (p \vee q) \wedge r$$

<u>p</u>	<u>q</u>	<u>r</u>	<u>$p \vee q$</u>	<u>$(p \vee q) \vee r$</u>	<u>$(p \vee q) \wedge r$</u>
T	T	T	T	T	T
T	T	F	T	T	F
T	F	T	T	T	T
T	F	F	T	T	F
F	T	T	T	T	T
F	T	F	T	T	F
F	F	T	F	T	F
F	F	F	F	F	F



$$(p \vee q) \wedge \neg r$$

p	q	r	$p \vee q$	$\neg r$	$(p \vee q) \wedge \neg r$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	F	F	F
F	F	F	F	T	F



$$(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$$

<u>p</u>	<u>q</u>	<u>$\neg p$</u>	<u>$q \rightarrow \neg p$</u>	<u>$p \leftrightarrow q$</u>	<u>$(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$</u>
T	T	F	F	T	F
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T



$$(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$$

<u>p</u>	<u>q</u>	<u>$\neg q$</u>	<u>$p \leftrightarrow q$</u>	<u>$p \leftrightarrow \neg q$</u>	<u>$(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$</u>
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	F	T	T
F	F	T	T	F	T



Summary of the lecture

- ▶ Introduction to the Course
- ▶ Propositions
- ▶ Logical Connectives
- ▶ Truth Tables
- ▶ Compound propositions



Supplementary Material & References

- ▶ **For propositions using truth tables, you can watch this video**



<https://study.com/academy/lesson/truth-table-definition-rules-examples.html>

- ▶ **Text-book**

Discrete Mathematics and Its Applications

Kenneth H. Rosen



Next Lecture

- ▶ 1.2 Applications of Propositional Logic
- ▶ 1.3 Propositional Equivalences

See you next lecture

